



Factors affecting the attitudes of principals of saudi general education public schools toward the inclusion of students with disabilities

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ABSTRACT

Background: Research has indicated that the attitudes of principals toward students with disabilities may affect the success of the implementation of inclusive programs in schools. This study was designed to address a gap in the research regarding Saudi principals' attitudes toward inclusion. Relationships between attitudes and a number of variables were analyzed (i.e., gender, years of experience, level of school, type of student disability, personal relationships with individuals with disabilities).

Methods and procedure: The study sample was drawn from the 600 public general education elementary, middle, and high schools with special education programs in Riyadh ($n = 366$ schools) and Jeddah ($n = 234$ schools) in Saudi Arabia. Each of the principals of these schools for AY2022/2023 was sent an email invitation to participate. A total of 403 respondents (67.17% response rate) successfully completed [Bailey's \(2004\)](#) Principals' Attitudes Toward Inclusive Education (PATIE) scale via electronic survey link.

Results and outcomes: Overall, the research found that Saudi principals of public general education schools have positive attitudes toward inclusion. The factors of training in special education, personal experience with individuals with disabilities, and years of experience as principal were found to be significant in regard to more positive attitudes. Conversely, the variables of gender, level of school, and type of student disability were not found to correlate with more or less positive attitudes.

Conclusions and implications: While the findings of the study bode well for the implementation of inclusion in Saudi public schools, certain limitations exist, including that the study sample was entirely drawn from the two largest cities in the country. In addition, the Saudi Ministry of Education has created an incentive plan that provides financial bonuses to principals at public schools who implement inclusive and special education programming in their schools. This indicates that the target population of this study might be more inclined to be positive toward inclusion than their colleagues at private schools. The findings have implications for expanding special education training in university teacher education programs and the creation of public awareness programs designed to improve understanding of disabilities and how inclusion positively impacts society as a whole.

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In schools that provide programs for students with disabilities—whether through full or partial inclusion—principals are critically influential elements to the success of these students and programs (DuFour & Mattos, 2013; Louis et al., 2010; Smith & Larwin, 2021). Principals are key to the team that supports students with disabilities. They also affect their schools in other important ways, including: (a) development and realization of the vision and goals of the school; (b) development of the capabilities of all school staff through professional development and other in-service training; (c) support of all students, regardless of disability status; and (d) through the promotion of a collaborative, inclusive, and positive environment in the school for staff and students (DuFour & Mattos, 2013; Gous et al., 2014; McLeskey et al., 2012; Smith & Larwin, 2021). In fact, through the appropriate engagement of such efforts, principals have a significant impact on the success of students in general and on inclusive education programs, specifically (Lyons et al., 2016). Therefore, principals' attitudes toward students with disabilities and the factors that affect these attitudes have been the subject of a number of research studies (e.g., Horrocks et al., 2008; Nguluma et al., 2017; Ramirez, 2006).

1. Principals' attitudes and key factors

Since the perceptions of principals are so important to the success of inclusion, several studies have examined the relationships between the attitudes of principals and certain variables (e.g., Boyle et al., 2013; Hadjidakou & Mnasonos, 2012; Nguluma et al., 2017; Praisner, 2003). However, despite the research attention the subject has received, the findings regarding correlations between these demographic factors and attitudes have been inconsistent. For example, while a study conducted in Cyprus by Hadjidakou and Mnasonos (2012) found male principals had more positive attitudes toward the inclusion of students with disabilities, other researchers found no such correlation between gender and attitudes (Horrocks et al., 2008; Praisner, 2003; Ramirez, 2006). Similarly, some research on teachers' attitudes found that the type of student disability may affect attitudes toward inclusion in the general education (GE) classroom (Aldosari, 2022). However, in the case of principals, researchers have not found strong correlations between type of disability and perceptions of inclusion (Abduljabbar & Masoud, 2002; Nguluma et al., 2017; Praisner, 2003).

Another oft-studied variable is that of training in special education (SE). On this factor, findings are again mixed with some research finding training correlates with more positive perceptions (Alshahrani, 2014; Avissar et al., 2003; Nguluma et al., 2017), and others finding that such training has no significant effect on attitudes (Conaway, 2018; Horrocks et al., 2008; Praisner, 2003). The causes of these variations must also be considered. For example, Graham and Spandagou (2011) asserted that negative attitudes of school principals toward inclusion are usually related to a lack of understanding regarding the requirements of successfully and appropriately implementing such programs. A study by Pazey and Cole (2013) found that many school principals report they lack training in how to administer SE programs. This becomes a particularly critical issue given the increasing diversity and numbers of students with special needs being served in GE classrooms.

In regard to length of experience, the findings are again inconsistent. While Avissar et al. (2003) found principals with more experience in administration are more negative toward inclusion, others found that more experience correlates with a more positive perception of inclusion (e.g., Conaway, 2018; Hadjidakou & Mnasonos, 2012; Horrocks et al., 2008; Nguluma et al., 2017). Specifically, in a study involving survey of both teachers and administrators, LeMay (2017) found that those respondents who indicated experience in the lower range (0–15 years) were more positive toward inclusive practices than those with greater experience (16–30+ years). In this study, those in administrative positions were found to be more positive toward inclusion than were teachers. This finding might have been related to the fact that the respondents in administrative positions also reported having more resources related to inclusion than did the teachers in the same study (LeMay, 2017). In contrast to these studies, other researchers have found that greater years of experience do not affect attitudes toward inclusion at all (e.g., Barnett & Monda-Amaya, 1998; Inzano, 1999; Ramirez, 2006).

Another variable that has received attention is that of the level of school, meaning elementary, intermediate (middle), or high school, for example. However, again, the findings on this factor vary from study to study. Certain research focused on students with autism spectrum disorder (ASD) found that elementary school principals' attitudes toward inclusion were more positive than those of their colleagues at the middle and high school levels (Horrocks et al., 2008). A study by Nguluma et al. (2017) conducted in Turkey found no significant differences in attitudes between principals of public elementary and middle schools regarding the inclusion of students with disabilities. Similarly, two studies that compared attitudes toward inclusion of high school and elementary school principals in the United States also found no significant differences between the two groups. Finally, an individual's personal experiences can often affect attitudes. On this factor, a number of studies have found a positive effect related to a principal's personal experience with individuals with disabilities outside of the school setting and perceptions of inclusion (Hadjidakou & Mnasonos, 2012 [Cyprus]; Nguluma et al., 2017 [Turkey]; Praisner, 2003 [United States]).

2. Special education in Saudi Arabia

There are both public and private schools in Saudi; international schools come under the "private" classification. At the preschool level, classes comprise both boys and girls who are taught solely by female teachers; at all levels above preschool, students are segregated by gender and taught by same-gender instructors. It is compulsory for all students to attend Grades 1–6. The public school curriculum at this primary stage is delivered in Arabic and although it is expanding, it is heavily centered on aspects of Islamic studies (Marie, 2024). It is free for all Saudi citizens (and certain others) to attend public schools for Grades 1–12. The curriculum at private primary schools with Arabic instruction is generally the same as that at public primary schools. However, these schools may offer instruction based "on a particular teaching standard or methodology" (Marie, 2024). Grades 7–12 at public schools, which in Saudi are comprised of "intermediate" and "secondary," tend to teach a broader array of subjects and included among these is typically the English language. Again, private schools differ in terms of focus on a certain methodology and/or having instruction presented in a

particular foreign language (English, German, etc.), as these international schools are for the children of expats (foreigners) from specific other countries. Due to the more diverse offerings and the costs associated with private/international schools (which can be substantial and out of the reach of the average Saudi citizen), there is some prestige associated with sending one's children to private schools in Saudi. Regardless of the type of school, some include programming for children with special needs and some do not.

Over the last 25 years or so, SE services in Saudi Arabia have evolved rapidly; however, true full inclusion has not yet been achieved. Beginning in 1999, there has been a strong push to implement at least mainstreaming programs in the country. Such programming involves having students with disabilities attend their local GE schools and being educated to varying degrees—typically dependent upon their type of disability—alongside their typical peers. In general, the Saudi Ministry of Education directs that students who have mild/moderate disabilities attend GE schools and be educated in the GE classroom with their typical peers for all or part of the day, receiving SE services in dedicated classrooms; those students with severe disabilities attend separate SE institutions (Al-Mousa, 1999; Ministry of Education of Saudi Arabia, 2008).

As with all education in Saudi Arabia, special education is segregated by gender, regardless of where it is received—in the GE classroom, in a dedicated SE classroom in a GE school, or at a dedicated SE institution. All mainstreaming programs attached to GE schools are under the management of the school principal, who is selected and appointed to the school by the Ministry of Education. Any candidate for a principal position must have worked as a teacher for a period of no less than 4 years and achieved performance appraisals of no less than “Excellent” in the most recent 2 years. However, the Ministry of Education does not require that principals possess an advanced degree (master's or higher).

3. Theoretical framework

This study is grounded in Burns's (1978) transformational leadership theory that asserts that the attitudes of leaders affect those of their employees (Balyer, 2012). This model is based on the idea that traits of the leader—i.e., attitude, strength, and vision—have significant effect throughout an organization, including by influencing other members to cooperate in achieving common goals (Tucker & Russell, 2004). The concept of transformational leadership can be applied to the role of administrator in the context of cooperating to achieve the common goals of the society by implementing successful programs to fulfill the requirements of the law and to fully support those with disabilities in the inclusive environment (Bass, 1985; Bass & Riggio, 2008). Providing education for the diverse population of individuals with disabilities benefits society as a whole, and all students, including those with disabilities, individually. However, change is often met with resistance in most organizations, so an understanding of transformational leadership practices is key. Other research has involved the application of transformational leadership to the specific environment of schools, including Balyer (2012) who investigated teachers' perceptions of transformational leadership.

4. Gap in the research

When principals view inclusion as a positive practice for all members of the school community, they are better able to drive the implementation of appropriate programs in their schools. However, past research—especially in Saudi Arabia—has tended to focus on teachers' attitudes toward inclusion rather than those of principals (e.g., Ahmed, 2015; Alasim & Paul, 2019; Aldosari, 2022; Alkhattabi et al., 2020; Alquraini, 2012). While such research is valuable because of the strong impact teachers have on all their students, the importance of principals' leadership role in schools cannot be ignored. This author was only able to locate three Saudi studies on the attitudes of principals (i.e., Abduljabbar & Masoud, 2002; Alothman, 2014; Alshahrani, 2014). Moreover, two of these studies, both of which are almost 10 years old, examined only one group within the total population of students with disabilities, that of the Deaf and Hard of Hearing (DHH; Alothman, 2014; Alshahrani, 2014). Furthermore, although the much older work by Abduljabbar and Masoud (2002) looked at principals of schools with broader populations of students with disabilities, they only examined the relationship between attitudes and two variables (i.e., principals' educational achievement and category of disability of student). Even outside of Saudi, there is a general lack of research focused solely on the attitudes of principals (Lindsey, 2009; Smith, 2011) and a lack of research that includes even small numbers of principals conducted in recent years.

Given the legislative and societal imperatives to implement inclusion in public schools, as well as the established importance of principals to achieve success in such programs, there is an urgent need to conduct more timely research on the attitudes of school principals toward inclusion. Additionally, such research must encompass a broader number of factors and determine their relevancy to principals' attitudes. For example, if factors such as professional development in special education correlate with more positive attitudes, then greater emphasis on such programs must be given.

Therefore, in this study I aimed to investigate the perceptions of Saudi principals in the cities of Riyadh and Jeddah regarding the inclusion of students with disabilities and to identify factors that may affect these perceptions. The objectives of the study were:

1. To examine the perceptions of Saudi principals of public general education schools regarding the inclusion of students with disabilities; and
2. To examine the relationship, if any, between certain variables (i.e., gender, years of experience as principal, personal relationship (s) with individuals with disabilities, training in special education, type of student disability, and school level) and principals' attitudes.

5. Methodology

In this study, I employed a quantitative method to systematically investigate the perceptions of Saudi principals' regarding the inclusion of students with disabilities. This approach allowed for the collection and analysis of numerical data to identify patterns and correlations between principals' attitudes and various demographic factors. Following the establishment of this framework, a survey was constructed to collect information on the perceptions of principals regarding the inclusion of students with disabilities in the GE school setting. Survey method was employed as its findings are considered more generalizable to a given target population (Creswell & Creswell, 2022). To conduct the main study and the pilot study, approval was obtained from the Institutional Review Board of [anonymized for review].

6. Setting

The study sample was drawn from public GE schools with SE programs during the 2022–2023 academic year in the two largest cities in Saudi Arabia, Riyadh and Jeddah. These two cities were selected because urban areas tend to have higher rates of SE programming within GE schools than less-populated settings. The capital city of Riyadh is in the central region; Jeddah is in the western region.

Jeddah has SE programs for students with disabilities in 234 GE public schools, including 102 elementary schools, 74 middle schools, and 58 high schools. Riyadh has a larger number of special education offerings spanning 366 schools; this number includes 194 elementary schools, 91 middle schools, and 81 high schools (Ministry of Education, 2021).

7. Participants

The population of interest for the current study were principals during AY2022–2023 of the 600 public GE schools with special education programs in the two cities. Contact information (i.e., email addresses) for all the principals of qualifying schools were obtained from the Ministry of Education. To conduct this study, all the principals of schools in Riyadh and Jeddah offering SE programming ($n = 600$) were invited via email to participate in an online survey. The email invitation noted that the survey link would be open for 3 weeks. The survey was opened on June 18, 2023 and closed on July 8, 2023. Of the total population of principals, 405 completed the survey; two of these responses were excluded due to missing data. Therefore, the final study sample comprised 403 respondents who provided usable data for statistical analysis, which is a 67.17% response rate. The demographic data of these 403 respondents is presented on Table 1.

8. Instrument

Prior to initiating the study and adapting the instrument, the researcher contacted Dr. Jeff Bailey, the author of the PATIE scale, for permission and direction regarding the implementation of the scale. After providing permission to adapt the scale for an Arabic-speaking study sample in Saudi Arabia, Dr. Bailey also provided instruction on how to score the instrument once the survey period had closed. This instrument has been utilized in a number of investigations evaluating principals' attitudes on the study topic in both the United States (Boston, 2017; Chandler, 2015; Lorio, 2011; Mahoney, 2018; Pennock, 2021) and other countries (e.g., Choi, 2008;

Table 1
Demographic Characteristics of the Public Elementary School Principals in the Study.

Variable		<i>n</i>	%
Gender	Male	224	55.58
	Female	179	44.42
Years of experience as principal	Less than 5 years	109	27.05
	5 – 10 years	181	44.91
	More than 10 years	113	28.04
Level of school	Elementary	223	55.33
	Middle school	102	25.31
	High school	78	19.35
Training in special education/inclusion	Yes	136	33.75
	No	267	66.25
Type of student disability	Intellectual disability	119	29.53
	Autism spectrum disorder	42	10.42
	Visual impairment	29	7.20
	Deaf/Hard of Hearing	44	10.92
	Learning disability	56	13.90
	Physical disability	93	23.08
	Multiple disabilities	20	4.96
Contact with individuals with disabilities	Yes	208	51.61
	No	195	48.39

Note. Percentages do not add up to 100.00% exactly due to rounding. The last variable, "Contact with individuals with disabilities," refers to personal contact, such as with friends, relatives, and colleagues.

Daodaoen et al., 2022; Nguluma et al., 2017; Sharma & Chow, 2008). In the present study, the PATIE scale was chosen because it employs a wide range of factors that can accurately measure information regarding principals' viewpoints.

The study survey comprised three sections. The first consisted of an explanation of the study's objectives, guidelines for responding to the questionnaire, and a consent form outlining the respondents' rights should they choose to participate. The second contained questions designed to obtain demographic information from the participants, including (a) gender, (b) years of experience as principal, (c) personal relationships, if any, with individuals with disabilities, (d) training in special education, (e) types of student disability encountered, and (f) school level. The third section presented the 24-item PATIE scale (Bailey, 2004), which is divided into five subscales:

- Factor 1, which includes five items related to Teacher Workload and Management (TWM);
- Factor 2, which includes six items related to Inclusion Benefits and Levels of Disability (IBLD);
- Factor 3, which includes seven items related to Learning Challenges in Inclusive Settings (LCIS);
- Factor 4, which includes three items related to Excluded Students (ES); and
- Factor 5, which includes three items related to Professional Training (PT).

Minor adjustments to the scale were made for this study, including the modification of some terms so as to be appropriate for the study population of Saudi principals. When investigating perceptions, a Likert scale is commonly employed, as they are efficient and highly accurate (Crano & Brewer, 2002; Leedy & Ormrod, 2001). Specifically, in this study a 5-point scale was utilized for the second part of the survey (5 = "strongly agree" to 1 = "strongly disagree").

8.1. Reliability and validity of the instrument

Different factors can affect the reliability of an instrument, including the internal consistency of the items of the scale and response bias. Cronbach's alpha, which ranges from $0.00 < \alpha < 1.00$, was used to assess the reliability of the scale. Cronbach's is broken down into the following categories: (a) not reliable: $0.00 \leq \alpha < 0.40$; (b) low reliability: $0.40 \leq \alpha < 0.60$; (c) reliable: $0.60 \leq \alpha < 0.80$; very reliable: $0.80 \leq \alpha < 1.00$ (Alpar, 2010). Bailey (2004) reported a Cronbach's alpha of .921 in his initial paper on the PATIE scale. Other researchers who have utilized the PATIE and validated it as part of their studies have also achieved reliable to very reliable Cronbach's alpha of 0.73 to 0.839 (Al-Ahmadi, 2009; Boston, 2017; Nguluma et al., 2017). This indicates that the PATIE achieves good reliability, significant accuracy, and validity of scoring (Taber, 2018).

In this type of survey, it is important to endeavor to counter response bias, which is when respondents answer survey questions in the way they think is the most appropriate or socially "correct" manner rather than based on how they actually feel. Therefore, the 24 items of the adapted PATIE scale included 14 negatively-phrased items and 10 positively-phrased items (Kreitchmann et al., 2019).

8.2. Translation

After permission was obtained from Dr. Bailey to use the PATIE scale for this study, the instrument was then translated into Arabic from the original English. The translation process went through several stages. Initially, the English version of the instrument was translated into Arabic by bilingual professionals in [anonymized for review] University's English Department, using the committee approach (Furukawa et al., 2014). Next, another bilingual person translated the instrument back into English without seeing the original English version. These procedures were followed to ensure the reliability and validity of the instrument. A pilot study was then conducted to assess the effectiveness and feasibility of the Arabic version of the PATIE scale, which explored the use of suitable language and the length of the survey. The final Arabic version of the PATIE scale considered the logical suggestions of all the respondents and the cultural relevance of the questions.

9. Pilot study

Given that this research involved a translation of a standard survey instrument, it was important to conduct a pilot study with a similar group of respondents to confirm the validity of the survey and to evaluate whether survey items had been correctly translated in a way that was accessible and appropriate for Arabic speakers. The respondents for the pilot study were obtained through the following process. Using the contact information from the Ministry of Education for the 600 public school principals who have SE programming in their public GE schools in Jeddah and Riyadh, the researcher personally contacted 65 principals in the two cities and asked each to participate in the survey for the pilot study. The request to participate in the survey for the purposes of the pilot study also included a description of the research and the purpose of the study. Of these 65 invited principals, 52 responded and accessed the link to take the survey. To prevent preset sensitization, these respondents to the pilot study were excluded from the main study (Crano & Brewer, 2002).

The Cronbach's alpha for the pilot study was 0.73. During the pilot study, some respondents identified Items 10 and 13 as ambiguous. Therefore, these items were examined to determine ambiguity and assessed for appropriate grammar and terminology that Arabic-speaking respondents could readily understand. After this evaluation and analysis, the survey to be administered to the main study sample was designed to incorporate and address the recommendations from the pilot study participants. These processes helped to calculate the time required for the participants to complete the survey, enhanced the quality of the survey content and the scale's validity, and corrected certain grammatical and technical errors in the final Arabic version of the PATIE scale.

10. Data analysis

Both descriptive and inferential statistics were used to analyze the data. To determine the factors associated with principals' perceptions of inclusion, one-way Analysis of Variance (ANOVA) and two-sample (or "independent samples") *t*-test were performed. Before analysis, the assumptions of homogeneity of variance and normal distribution were checked and met for both statistical analyses. Two-sample *t*-test was used to compare the significant differences between two groups (e.g., gender and another variable). ANOVA was used to compare more than two variables as it is considered appropriate for use when determining if there are statistically significant differences between two independent groups based on a continuous outcome variable. Both independent *t*-test and ANOVA were used to analyze differences in attitudes toward inclusion. Before conducting these analyses, I confirmed that the data met the necessary assumptions required to conduct each statistical test. These are that: (a) "data values must be independent"; (b) "data ... are normally distributed"; and (c) there is homogeneity of variance (Statistics Knowledge Portal, 2024).

Normality test for variables were conducted to affirm the normal distribution. The homogeneity of variance test was conducted using Levene's test as provided in IBM SPSS (Version 27). Next, Cohen's *d* effect size was calculated to measure the magnitude of difference. Effect sizes are important to understanding the magnitude of differences found in the analysis, providing a more comprehensive understanding of the practical significance of the findings. Reporting effect sizes complements the statistical significance indicated by *p*-values, as this offers insights into the real-world implications of the differences observed in the study.

11. Results

Skewness and kurtosis values for all 24 items of the survey were calculated to assess the data normality. The skewness values ranged from -0.822 to -1.347 and the kurtosis values ranged from 0.106 to -0.922 for all 24 items. This means that all were within the ± 2 threshold. Given these findings, it can be inferred that the data distribution did not deviate significantly from normality. This means the data is symmetrical enough and does not have excessively heavy or light tails, making it suitable for further analysis with parametric tests. The acceptance of the normality assumption based on these skewness and kurtosis values supports the appropriateness of using two-sample *t*-test and one-way ANOVA in the study.

The 403 respondents' scores had a mean of 3.61 ($SD = 0.75$), which puts the perceptions of these principals in the $3.41 - 4.2$ range. This indicates an overall slightly positive attitude toward the inclusion of students with disabilities (see Table 2). The survey responses of each principal were analyzed using two-sample *t*-test, one-way ANOVA, and Scheffé Test.

12. Results of two-sample *t*-tests

Two-sample *t*-test was used to determine whether three variables are a factor in predicting principals' attitudes: gender, training in inclusion, and contact with individuals with disabilities (familiarity) in personal lives. The results of these tests are presented on Table 3. Regarding gender, no significant differences ($t(402) = -0.569$, $p > 0.05$) were found between the perceptions of male respondents ($n = 224$; $M = 3.59$, $SD = 0.77$) and those of female respondents ($n = 179$; $M = 3.63$, $SD = 0.72$). The corresponding effect size was found to be low (Cohen's $d = 0.053$).

However, statistically significant differences ($t(402) = 2.098$, $p < 0.05$) were found in respondents' perceptions regarding inclusion based on whether they had received ($n = 136$; $M = 3.72$, $SD = 0.74$) or had not received ($n = 267$; $M = 3.55$, $SD = 0.75$) training in inclusive education. These results indicated that the average positive perceptions were significantly higher for participants with training in inclusive education than for those without it. Here, Cohen's d effect size was calculated to be 0.23 .

Moreover, the results of the two-sample *t*-test on the variable of personal contact with individuals with disabilities found statistically significant differences ($t(402) = 2.666$, $p < 0.05$) between those respondents who reported not having such contact ($n = 195$; $M = 3.51$, $SD = 0.77$) and those that reported that they had personal relationships with individuals with disabilities outside of school ($n = 208$; $M = 3.70$, $SD = 0.71$). The calculated effect size here was 0.26 , which indicates a significant difference between the two groups. In this case, average positive perceptions were significantly higher for respondents who stated they had personal contact outside of school with people with disabilities (see Fig. 1).

13. Results of one-way ANOVA analyses

One-way ANOVA was conducted with three of the factors, which were years of experience, level of school administered (primary, intermediate, high school), and type of student disability. On level of school, no statistically significant differences were found in

Table 2
Interpretation of Average Mean Scores of Perceptions Toward Inclusion on PATIE Scale.

Scale	Score Range	Mean Range
5	5.00 – 4.21	Strongly agree
4	4.20 – 3.41	Agree
3	3.40 – 2.61	Neutral
2	2.60 – 1.81	Disagree
1	1.80 – 1.00	Strongly disagree

Table 3
Results of the Two-Sample t-Tests.

Variable	N	M	SD	t-value	p
Gender	224	3.59	0.77	− 0.569	0.570
Male	179	3.63	0.72		
Female					
Special education training	136	3.72	0.74	2.098	0.037
Yes	267	3.55	0.75		
No					
Contact w/those w/disabilities	208	3.70	0.71	2.666	0.008
	195	3.51	0.77		

Note. The third variable regarding contact relates to the respondent's familiarity with people with disabilities in their private life outside of school.

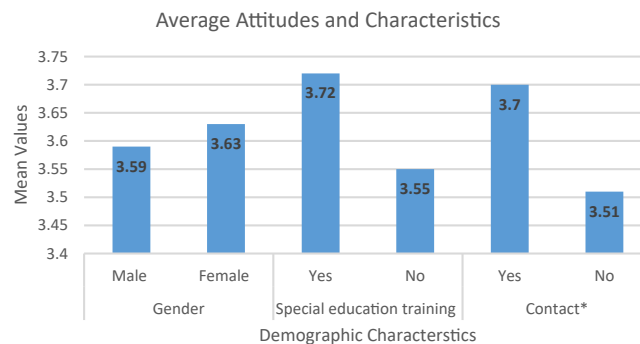


Fig. 1. Principals' Attitudes Toward Inclusion as Related to Demographic Characteristics. Note. The y-axis indicates the mean values of the principals' attitudes toward inclusion. *The category of "Contact" refers to respondent's familiarity with people with disabilities in private life outside of school.

respondents' perceptions toward inclusion related to the level of the school they administered ($F(2, 400) = 1.36, p = 0.258 > 0.05$). Similarly, no statistically significant differences in perceptions were found related to the type of student disability ($F(6, 396) = 1.48, p = 0.185 > 0.05$). In contrast, I found statistically significant differences in attitudes of principals toward the inclusion of students with disabilities based on their years of experience as principal ($F(2, 400) = 11.99, p < 0.05$). These findings are presented on Table 4.

14. Results of Scheffé test

It is sometimes useful "to make unplanned comparisons" following ANOVA (Kenton, 2021). Therefore, using the post hoc Scheffé Test, a multiple comparison was run on the factor of years of experience in the role of principal. Comparisons were made between each of the three ranges of years of experience and the other two ranges. Based on the means presented on Table 5 and the Scheffé Test (see Fig. 2), less than 5 years of experience as principals was found to correlate with more positive perceptions toward inclusion than did 5 – 10 years of experience ($p < 0.05$) and more than 10 years of experience ($p < 0.05$). Moreover, principals with 5–10 years of experience were found to have more positive perceptions of inclusion than those with more than 10 years of experience

Table 4
Results of One-Way ANOVA.

Variable	SS	Df	MS	F-value	p
Years of experience	12.69	2	6.35	11.989	0.000
Between groups	211.70	400	0.53		
Within groups	224.39	402			
Total					
Level of School	1.52	2	0.76	1.360	0.258
Between groups	222.87	400	0.56		
Within groups	224.39	402			
Total					
Type of Student Disability	4.91	6	0.82	1.477	0.185
Between groups	219.48	396	0.55		
Within groups	224.39	402			
Total					

Note. N = 403. SS = Sum of Squares; MS = Mean Square.
 $p < 0.05$.

Table 5
Descriptive Statistics of Years of Experience as Principals.

Category	n	M	SD	SE	95% Confidence Interval for Mean		Min.	Max.
					Lower Bound	Upper Bound		
< 5 years	109	3.85	0.64	0.061	3.73	3.97	1.70	4.87
5 – 10 years	181	3.61	0.70	0.052	3.51	3.71	1.95	4.93
> 10 years	113	3.37	0.85	0.080	3.21	3.53	1.70	4.91
TOTAL	403	3.61	0.75	0.037	3.54	3.68	1.70	4.93

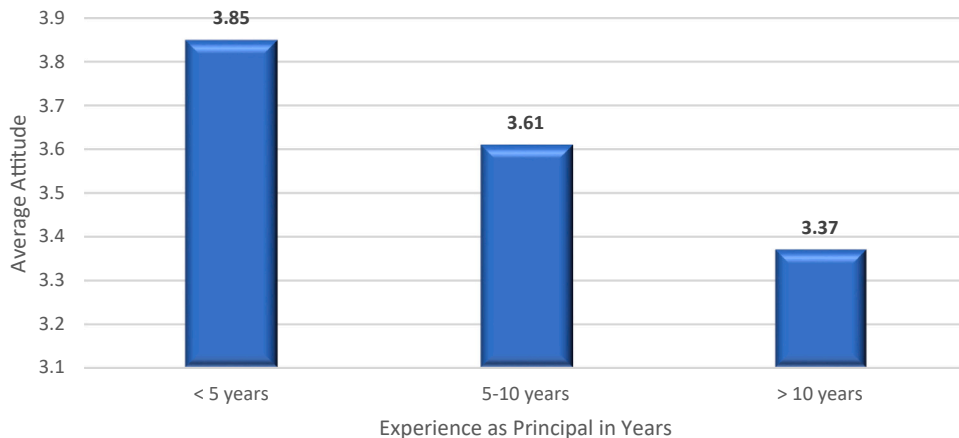


Fig. 2. Years of Experience as Principal Multiple Comparison Test. Note. This figure presents the data on the respondents' averaged attitudes toward inclusion as related to their years of experience as principal.

($p < 0.05$).

15. Discussion

This study analyzed whether the attitudes of principals of public GE schools in Riyadh and Jeddah toward the inclusion of students with disabilities were related to certain identified factors. The relationships between factors and perceptions varied, but respondents' overall attitudes toward inclusion were found to be slightly favorable. This result is consistent with those of a number of studies that found that in general, school principals have positive attitudes toward the integration of students with disabilities into the general education setting so that these children can be educated in the least restrictive environment with their peers with and without disabilities (Avissar et al., 2003; Chandler, 2015; Horrocks et al., 2008; Nguluma et al., 2017). However, unlike in this research, some other studies have found principals to be negative toward inclusion (e.g., Ball & Green, 2014; Muega, 2016).

It is possible that the findings of this study are affected by certain aspects of the Saudi educational system. For example, the Ministry of Education has implemented a bonus program that provides financial incentives to principals of schools that include SE programming for students with disabilities. Specifically, according to Circular No. 262/27 (dated 2/19/2001), principals of GE schools with such programs receive a 20% increase in monthly salary (Ministry of Education, 2013). In addition, although principals are integral to setting policy, implementing inclusive programming, and motivating staff to achieve different goals, principals do not typically have direct contact with students with disabilities. As a result, principals may not understand as well as teachers the day-to-day challenges related to ensuring that these students achieve to their potential (Avissar et al., 2003; Cook et al., 1999).

16. Discussion of the findings on the different variables

16.1. Gender

The results of the current study found no difference between the attitudes of male and female principals toward the inclusion of students with disabilities in the GE classroom. This might be related to the fact that all these male and female principals work under the umbrella of one ministry and one education department. In addition, they have the same salaries and benefits and the same responsibilities and tasks. Moreover, all these principals received the same preparation and have the same qualifications, graduated from the same set of universities, and went through the same stages of career progression before becoming school principals. This result is consistent with the results of a number of previous studies, which found no significant differences in the attitudes of male and female school principals toward the inclusion of students with disabilities (e.g., Chandler et al., 2015; Horrocks et al., 2008; Nguluma et al., 2017; Praisner, 2003; Ramirez, 2006; Ugwu & Onukwufor, 2018). However, Hadjidakou and Mnasonos (2012) found that female

principals are more positive toward inclusion. Similarly, although Boyle et al. (2013) found that teachers in general are positive toward inclusion, they identified female teachers as being much more positive than males.

16.2. Number of years as principal

Regarding years as principal, the current study's findings indicate that those with less experience in the position of principal are more positive. A possible explanation for this is that principals with less experience are typically more recently graduated from university and have therefore received more progressive training regarding inclusion than their colleagues with longer experience. In the last 20 years, Saudi SE policies and practices have undergone substantial changes, which are reflected in university offerings. University education departments now include a number of optional and compulsory courses in special education. This means that newly graduated teachers, from whom future principals will be chosen, are more knowledgeable about special education than older, more experienced teachers and principals.

Principals who are relatively new to their roles also might not be as entrenched in traditional educational models, making them more adaptable to the adoption of inclusive practices. In addition, it is also possible that less experienced principals are also in the demographic of more recently graduated teachers and therefore were more recently exposed to current training and educational curricula that emphasize the importance and benefits of inclusive education. This recent exposure could influence their attitudes, making them more positive toward inclusion compared to more experienced principals who may have received their training when inclusive education was less accepted. In addition, less experienced principals may have greater motivation to implement change so as to have a significant impact on their schools. This drive also could translate into more positive attitudes toward inclusion as a means to enhance educational outcomes and equity.

This result of the current study differs from the findings of several studies that found that more experience translates to more positive perceptions of inclusion (e.g., Conaway, 2018; Hadjidakou & Mnasonos, 2012; Horrocks et al., 2008; Nguluma et al., 2017). These findings also conflict with those of Barnett and Monda-Amaya (1998), Inzano (1999), and Ramirez (2006), who found no relationship at all between number of years in administration and attitudes.

16.3. Type of student disability

Although the type of disability has been correlated with attitudes of educators toward inclusion (Aldosari, 2022), the current study did not find any effect related to disability type. This result is consistent with those of Abduljabbar and Masoud (2002) and Praisner (2003), who also did not find any strong correlation between type of disability and principals' perceptions of inclusion. It is difficult to determine why findings on this variable might differ from study to study. However, it is possible that other studies differed in terms of the range of disability type included in the particular school as compared to the range of disability type included in the schools involved in the current study.

16.4. Level of school

Regarding the level of the school where the principal was assigned, again the current study found no correlation with attitudes. This result is consistent with that of Nguluma et al. (2017), who found that school principals of elementary and middle schools in Turkey did not differ significantly in how they view students with disabilities and the importance of their inclusion in public GE schools. Similarly, Geter (1998) and Maricle (2001) also found no significant differences between the attitudes of U.S. principals of high schools and those of elementary schools. In contrast, Horrocks et al. (2008) found that principals at elementary schools have more favorable attitudes toward inclusion than those at middle and high schools. One possible explanation for the absence of differences in the attitudes of school principals toward inclusion related to school level might be due to the fact that the responsibility of managing programs for students with disabilities at public GE schools in Saudi are primarily the responsibility of district supervisors and SE teachers. The principals mostly have only limited interactions with students with disabilities. Therefore, they might not be aware of the variations in support and accommodation that students with different types of disabilities require as compared to others more directly involved in the education of these students (Avissar et al., 2003; Cook et al., 1999).

16.5. Personal relationships with individuals with disabilities

The current study found a relationship between principals' direct personal experience with individuals with disabilities in their lives outside of school and their perceptions of inclusion. Those with such personal experience were found to be significantly more positive toward students with disabilities in the GE school setting. Such personal relationships can serve as a powerful motivator for principals to pursue professional development opportunities related to special education and inclusion. This may further enhance their positive attitudes. These principals might also be more inclined to advocate for inclusive practices within their schools, which in turn influences their school's culture and the attitudes of other staff members toward inclusion. In Nguluma et al. (2017), the authors also found that personal relationships with those with disabilities positively affect attitudes. The finding of the current study was also consistent with those of Hadjidakou and Mnasonos (2012) in Cyprus and of Praisner (2003) in the United States.

16.6. Training in inclusive education

As with past studies, the current study also found that the principals who had received training in special education had more positive attitudes toward inclusion. When individuals have knowledge of disabilities, an understanding of the laws on inclusion/special education, and the overall benefits of inclusion to students and society, they tend to be more positive toward the practice (Ramirez, 2006). Notably, although Ngwokabuenui (2013) did not find SE training to be a factor, the author did find the separate variable in that study of knowledge of SE law to have a significant positive effect on principals' attitudes toward inclusion. This difference might be due to what comprises such training in Cameroon, which was the setting for Ngwokabuenui (2013), versus how it is presented in other countries, including Saudi Arabia. The finding on training of the current study is also consistent with that of Chandler (2015), Praisner (2003), and Vazquez (2010). However, a few studies have found that formal training in special education has no appreciable impact on the attitudes of principals toward inclusion (e.g., Conaway, 2018; Horrocks et al., 2008); and one study, that of Ball and Green (2014), found a negative relationship between training and principals' attitudes toward inclusion.

17. Implications

The findings of this study have several important implications for future policies and practices in Saudi Arabian education in regard to inclusion. For example, the findings indicate that efforts to enhance professional development for Saudi principals on the topic of inclusive education can foster positive attitudes toward inclusion. The results also highlight the potential of personal experience with individuals with disabilities to shape more progressive attitudes toward inclusion. This suggests that there is value in programs that raise awareness and engagement. Furthermore, the findings support the continuation and refinement of incentive schemes for principals who effectively implement inclusive education programs. The insights developed through this research can inform future educational policies and practices, emphasizing the importance of inclusive leadership training, experience-based understanding, and targeted incentives to improve the inclusion of students with disabilities in Saudi schools.

When principals have positive attitudes toward inclusion, they are more likely to understand the goals of such practices as well as the necessary steps that must be taken to make inclusive education successful. This can also lead to a better understanding of how to best define the roles of each member of the Individualized Education Plan (IEP) team. Attitudes also affect principals' understanding of the necessary budgetary expenditures required to successfully fund inclusive education in order to create an educational environment that fosters cooperation, belonging, and acceptance among all school staff and students (Chandler, 2015; Sari, 2007). Furthermore, it is a continuous challenge for teacher preparation programs to rethink practices and revamp strategies in order to prepare future educators with the skills they require to meet the needs of diverse student bodies. Researchers have also found that principals state that they lack understanding of and require additional training in special education and in the administration of SE programs (Graham & Spandagou, 2011; Pazey & Cole, 2013). Findings like these indicate that as Saudi Arabia moves toward more inclusive education for students with disabilities, it would be beneficial for the Saudi Ministry of Education to direct that university teacher education curricula be expanded in terms of requiring that future teachers—who are also potential future principals—be required to complete more coursework on not just special education but on all aspects of successfully implementing and administering SE programs.

Moreover, based on the findings herein, I recommend the Ministry of Education offer in-service principals training and professional development that focuses on the theory of metacognition, based on Drigas and Mitsea's (2021) model of metacognition training. This type of program may help principals adjust their attitudes toward students with disabilities by educating them in how to evaluate their own biases and preconceptions regarding the population of students with disabilities. This could assist in-service principals in more easily perceiving the clear benefits of inclusive programming (Drigas & Mitsea, 2020).

18. Limitations

The current study had certain limitations. These include the method by which the study sample was obtained and certain characteristics of the sample, both of which are related to the fact that the participants were recruited from just two major cities in Saudi Arabia (Riyadh and Jeddah). As a result, the findings might not be generalizable to rural cities in the country or even other major Saudi cities with different demographics. Moreover, public school staff might have different experiences than their peers at private schools due to different levels of funding and job security. In addition, the use of the PATIE might be a limitation as researchers are advised to exercise caution regarding the interpretation of research findings when survey method is employed (Munn & Drever, 1999). It is also possible that research bias (conscious or unconscious) might exist in the study sample, since the email invitation to participate included an explanation of the study objectives (Kreitchmann et al., 2019).

19. Recommendations for future research

There is a general lack of research regarding the perceptions of administrators about inclusion, especially with principals, vice principals, and supervisors of SE programs. Throughout this study's literature review, this concern was found to be particularly notable in the context of Saudi Arabia. Therefore, future studies should be conducted to investigate the attitudes of these different types of administrators not just in other major cities and geographically different (e.g., rural) areas of the country, but also with larger groups of participants to obtain more data.

Given the previously-noted narrow focus of the current study regarding the population surveyed, future studies also might investigate the same topic using the PATIE with principals and other administrators at Saudi kindergartens. Furthermore, future

studies might implement the PATIE scale in other Arab countries to obtain more data regarding principals' attitudes across the region and whether attitudes vary among the different countries. Expanding the research to these other areas will supplement and strengthen our understanding of these administrators' perceptions of inclusion. It may also allow us to better understand the implications of such attitudes, the factors that influence them, and the ways in which students with disabilities can be successfully supported in Saudi schools.

CRediT authorship contribution statement

Mubarak S. Aldosari: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Data availability

Data will be made available on request.

[PATIE Scale \(Original data\)](#) (PATIE Scale)

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